

### SCHEMATIC (see Page 2)

### FET NONLINEAR MODEL PARAMETERS <sup>(1)</sup>

| Parameters | Q1       | Parameters | Q1   |
|------------|----------|------------|------|
| VTO        | -0.85    | RG         | 4    |
| VTOSC      | 0        | RD         | 4    |
| ALPHA      | 6        | RS         | 3    |
| BETA       | 0.11     | RGMET      | 0    |
| TQGAMMA    | 0.07     | TNOM       | 27   |
| TQGAMMAAC  | 0.087    | XTI        | 3    |
| Q          | 2        | EG         | 1.43 |
| TQDELTA    | 0.4      | VTOTC      | 0    |
| VBI        | 0.715    | BETATCE    | 0    |
| IS         | 3e-13    | FFE        | 1    |
| N          | 1.22     |            |      |
| RIS        | 0        |            |      |
| RID        | 0        |            |      |
| TAU        | 5e-12    |            |      |
| CDS        | 0.17e-12 |            |      |
| RDB        | 1000     |            |      |
| CBS        | 1e-9     |            |      |
| CGS        | 0.3e-12  |            |      |
| CGD        | 0.02e-12 |            |      |
| DELTA1     | 0.3      |            |      |
| DELTA2     | 0.4      |            |      |
| FC         | 0.5      |            |      |
| VBR        | Infinity |            |      |

(1) ADS TOM Model

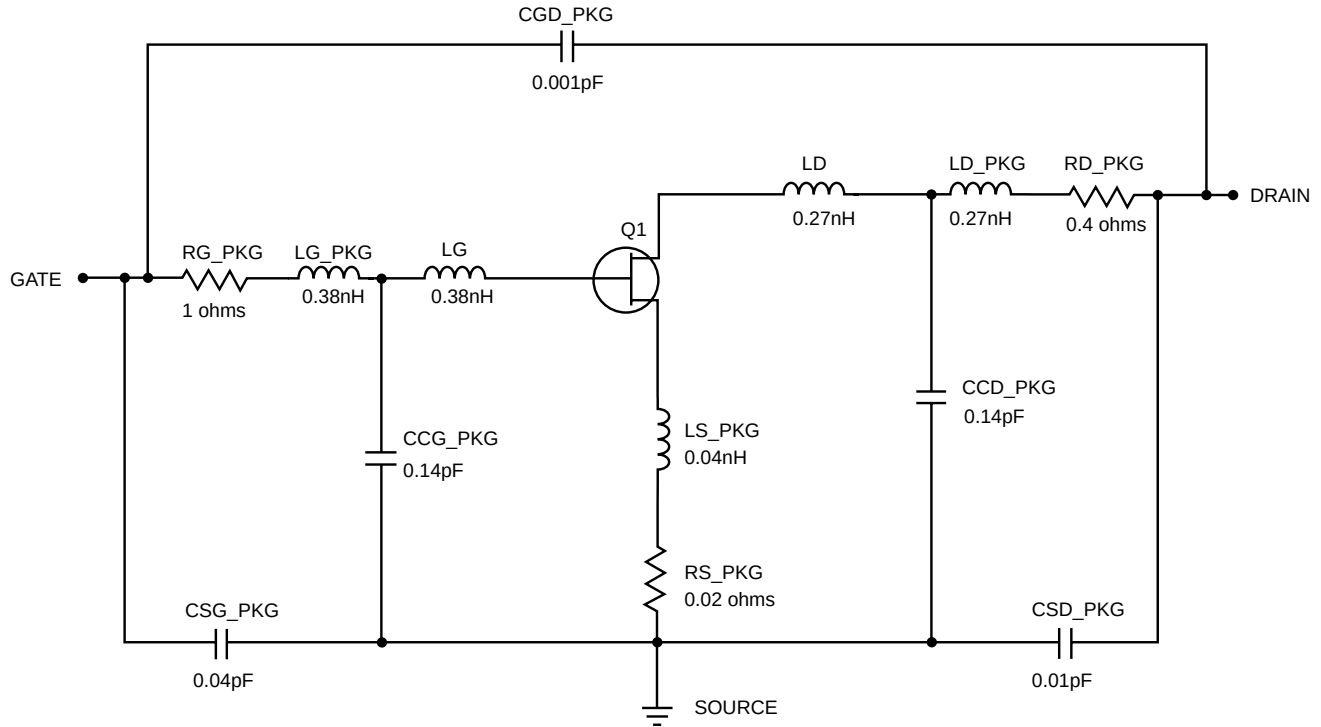
### UNITS

| Parameter   | Units   |
|-------------|---------|
| time        | seconds |
| capacitance | farads  |
| inductance  | henries |
| resistance  | ohms    |
| voltage     | volts   |
| current     | amps    |

### MODEL RANGE

Frequency: 0.1 to 18 GHz  
 Bias:  $V_{DS} = 1\text{ V to }3\text{ V}$ ,  $I_D = 5\text{ mA to }30\text{ mA}$   
 Date: 3/2002

## SCHEMATIC



## Life Support Applications

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